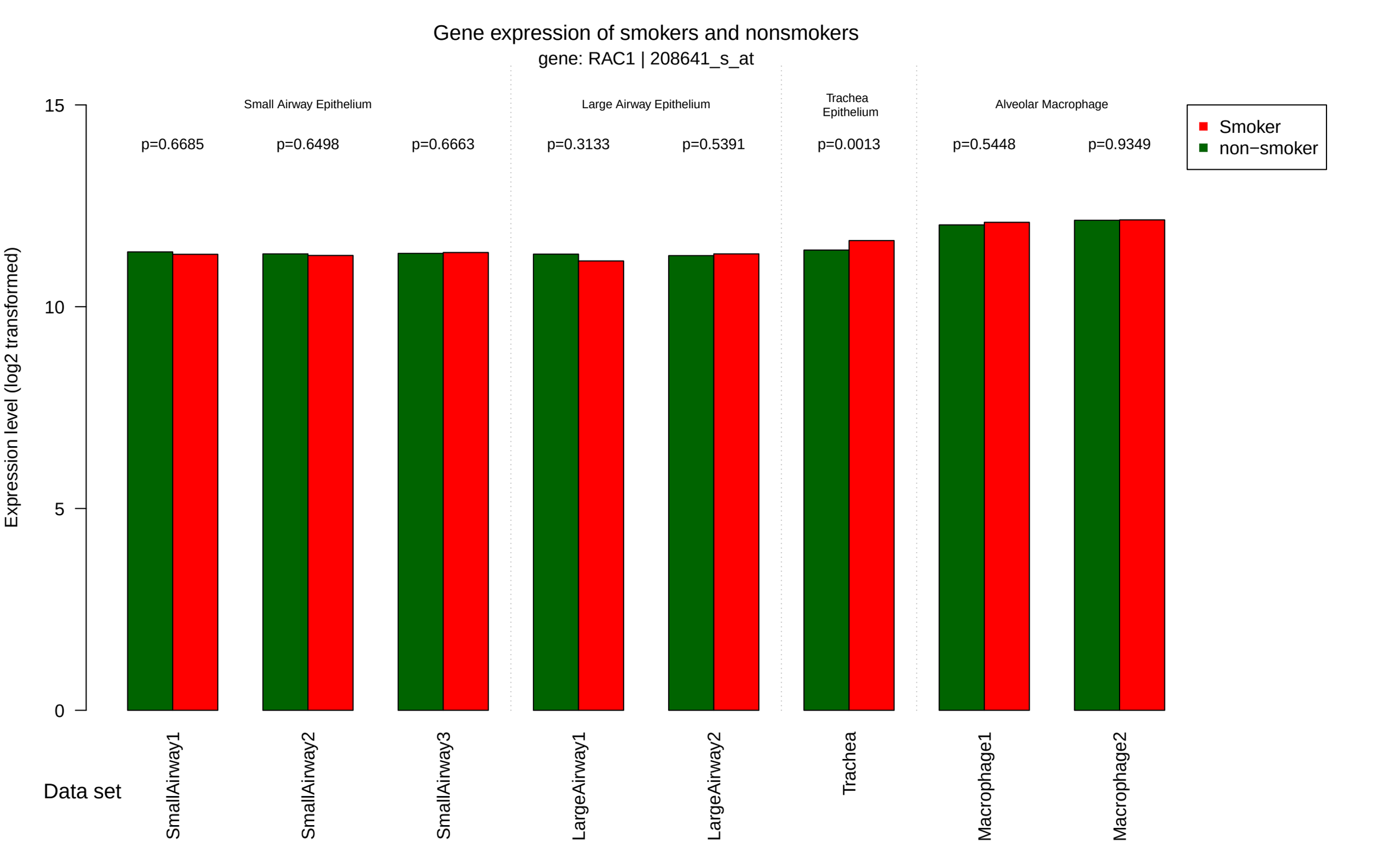
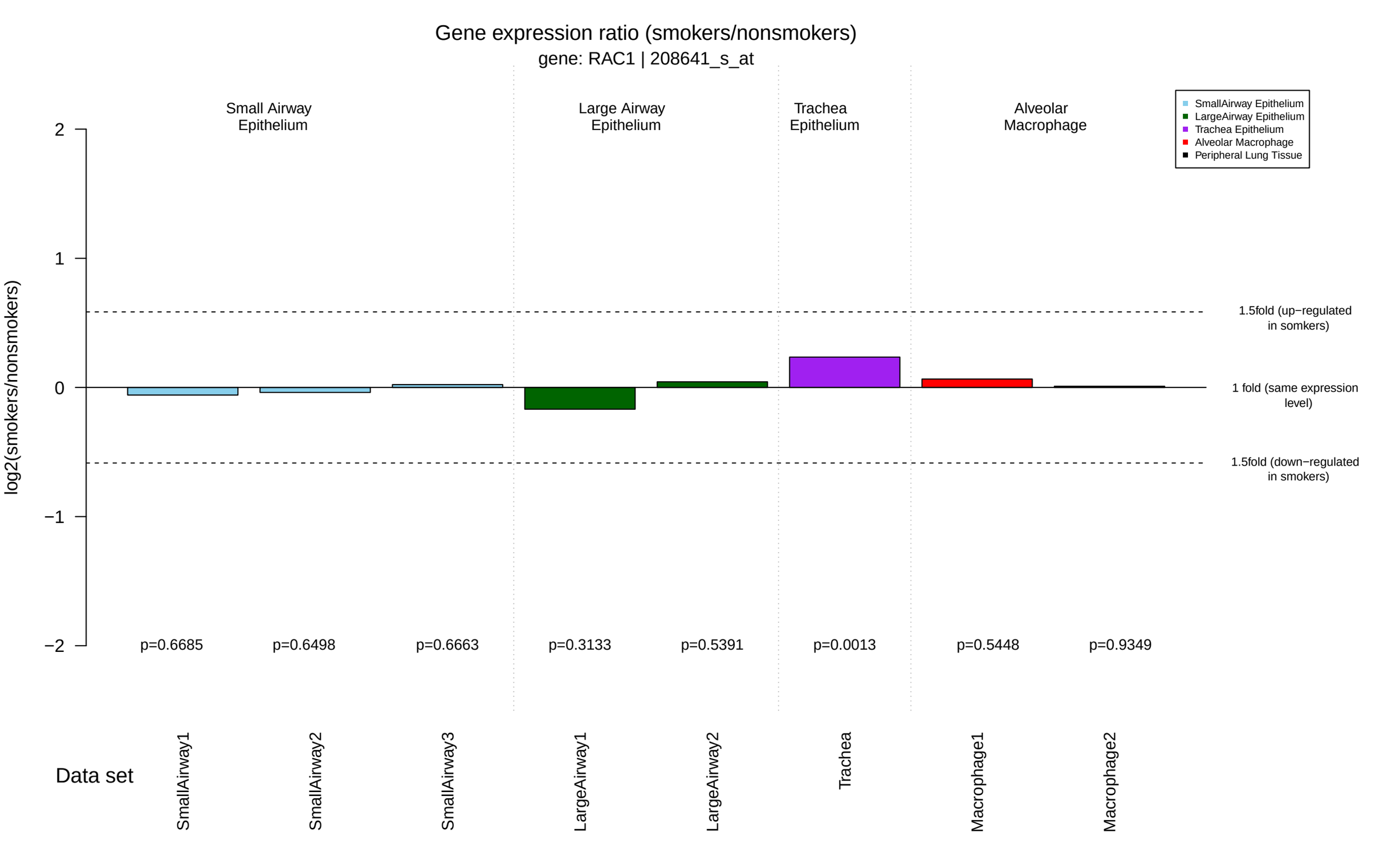
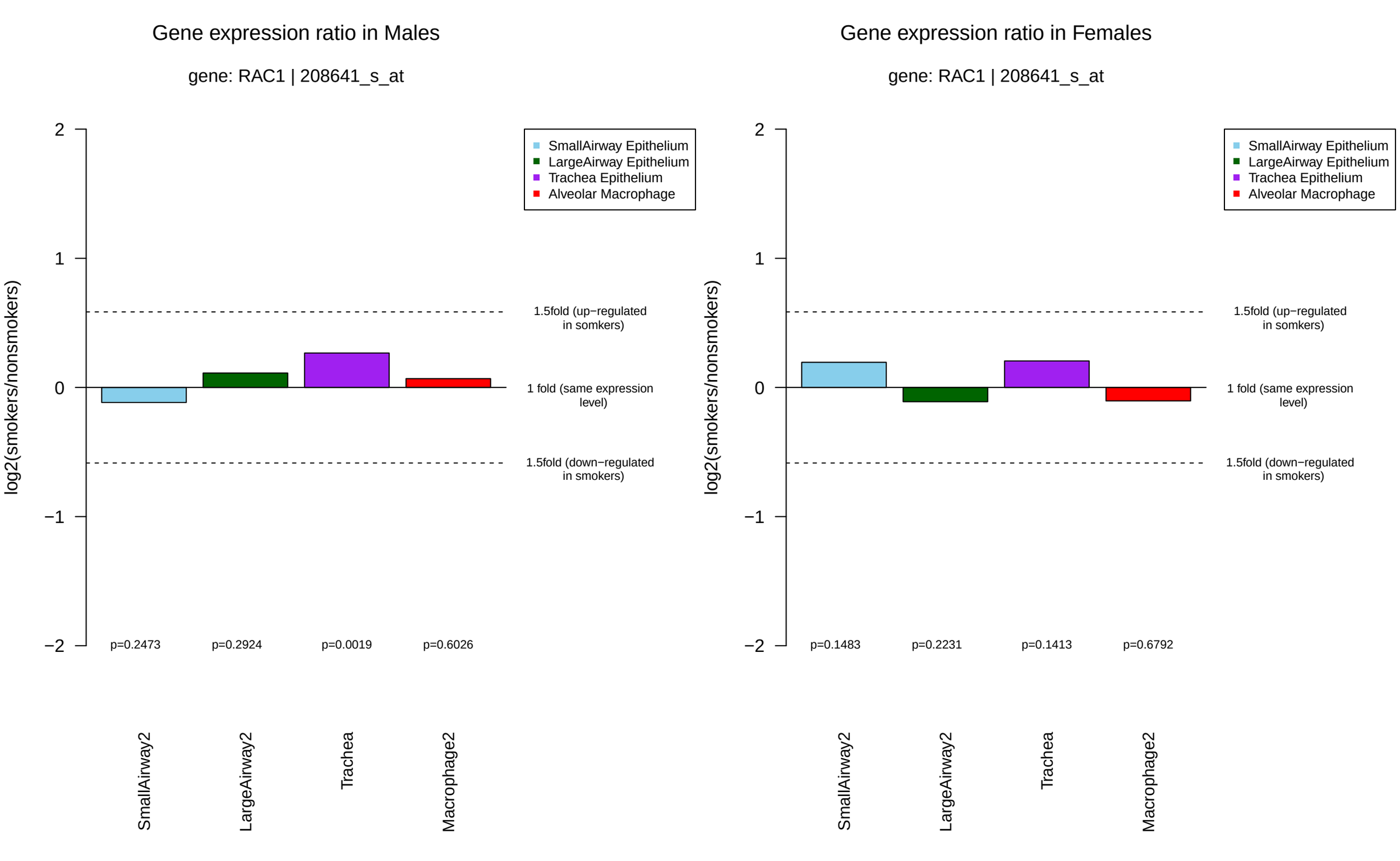
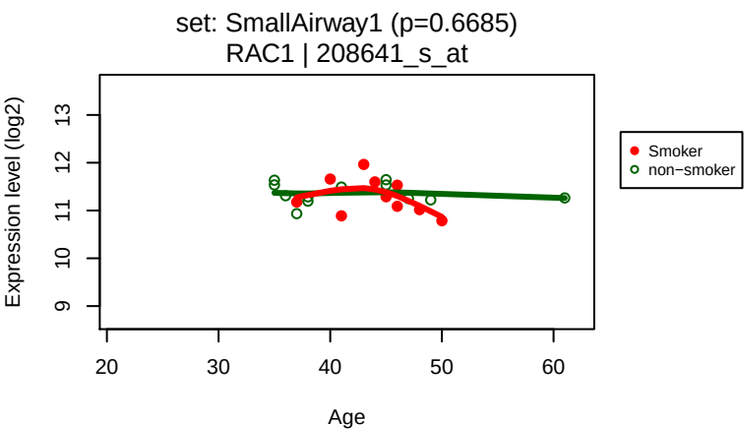
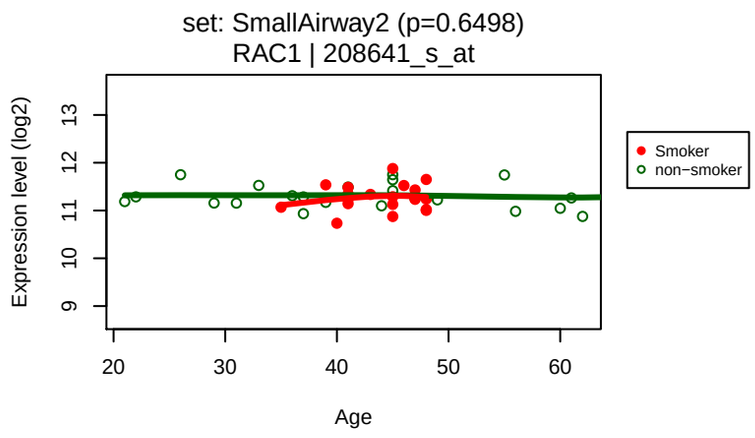
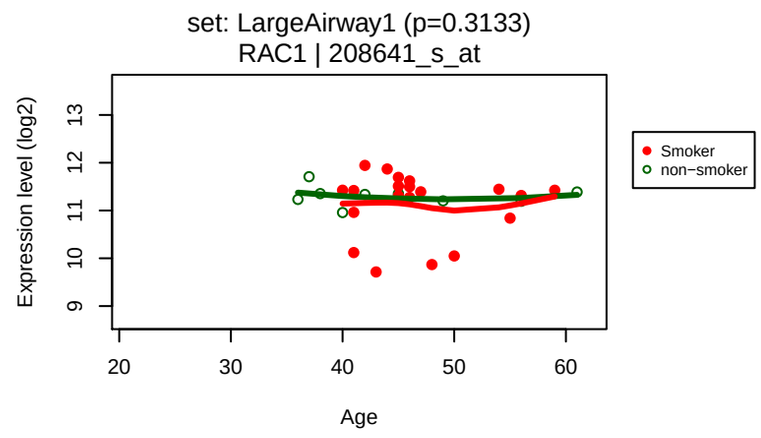

Gene expression of smokers and nonsmokers
gene: RAC1 | 208641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6685
p=0.6498
p=0.6663
p=0.3133
p=0.5391
p=0.0013
p=0.5448
p=0.9349
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

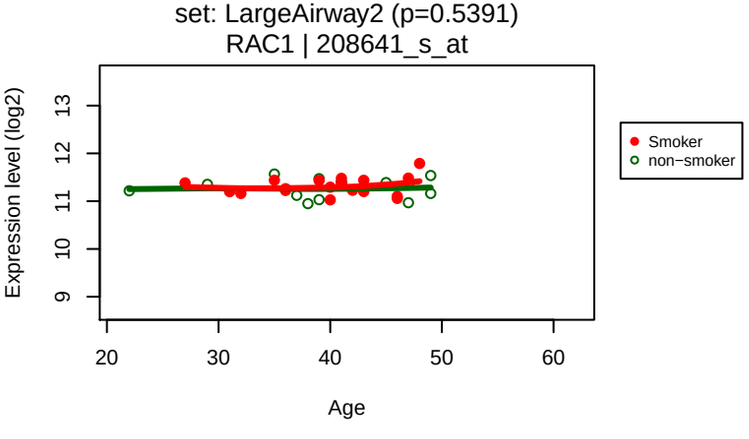

Gene expression ratio (smokers/nonsmokers)
gene: RAC1 | 208641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6685
p=0.6498
p=0.6663
p=0.3133
p=0.5391
p=0.0013
p=0.5448
p=0.9349
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

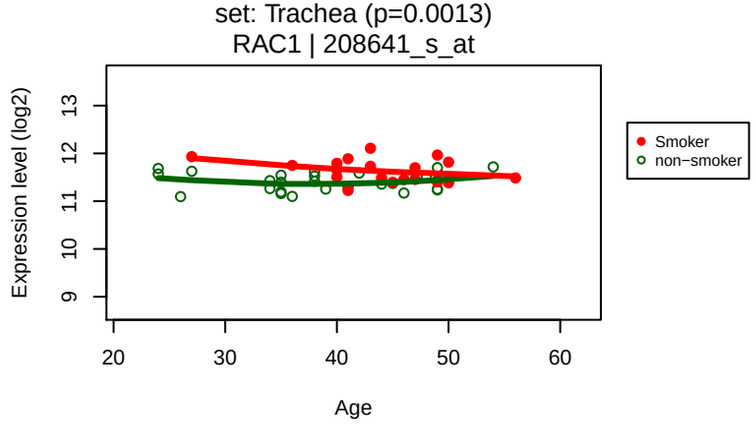

Gene expression ratio in Males
gene: RAC1 | 208641_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2473
p=0.2924
p=0.0019
p=0.6026
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC1 | 208641_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1483
p=0.2231
p=0.1413
p=0.6792
SmallAirway2
LargeAirway2
Trachea
Macrophage2

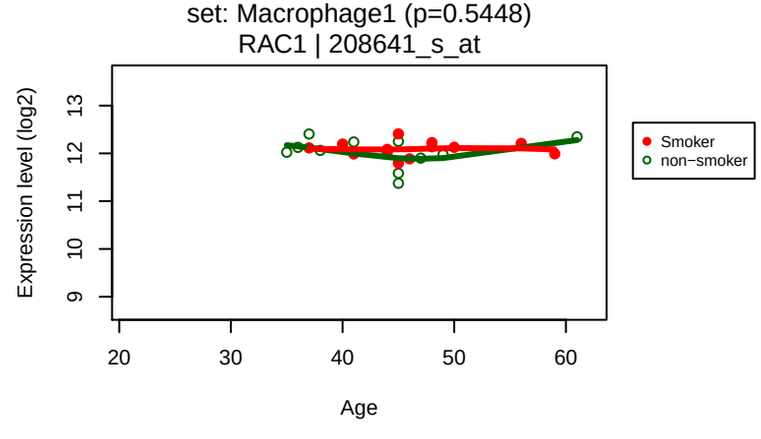

set: SmallAirway1 (p=0.6685)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker

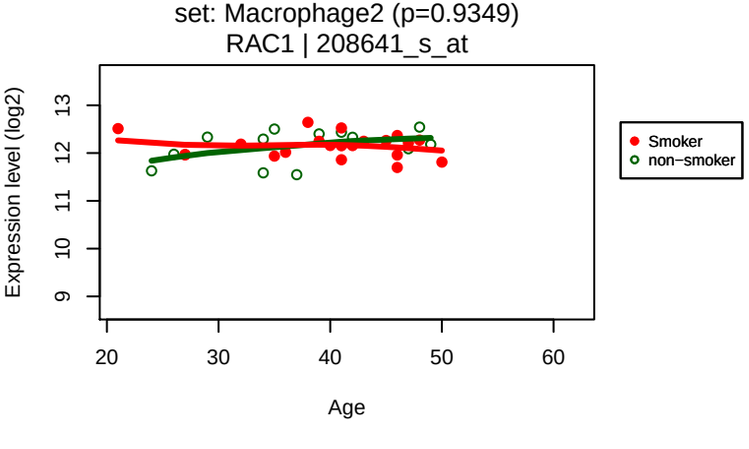

set: SmallAirway2 (p=0.6498)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker

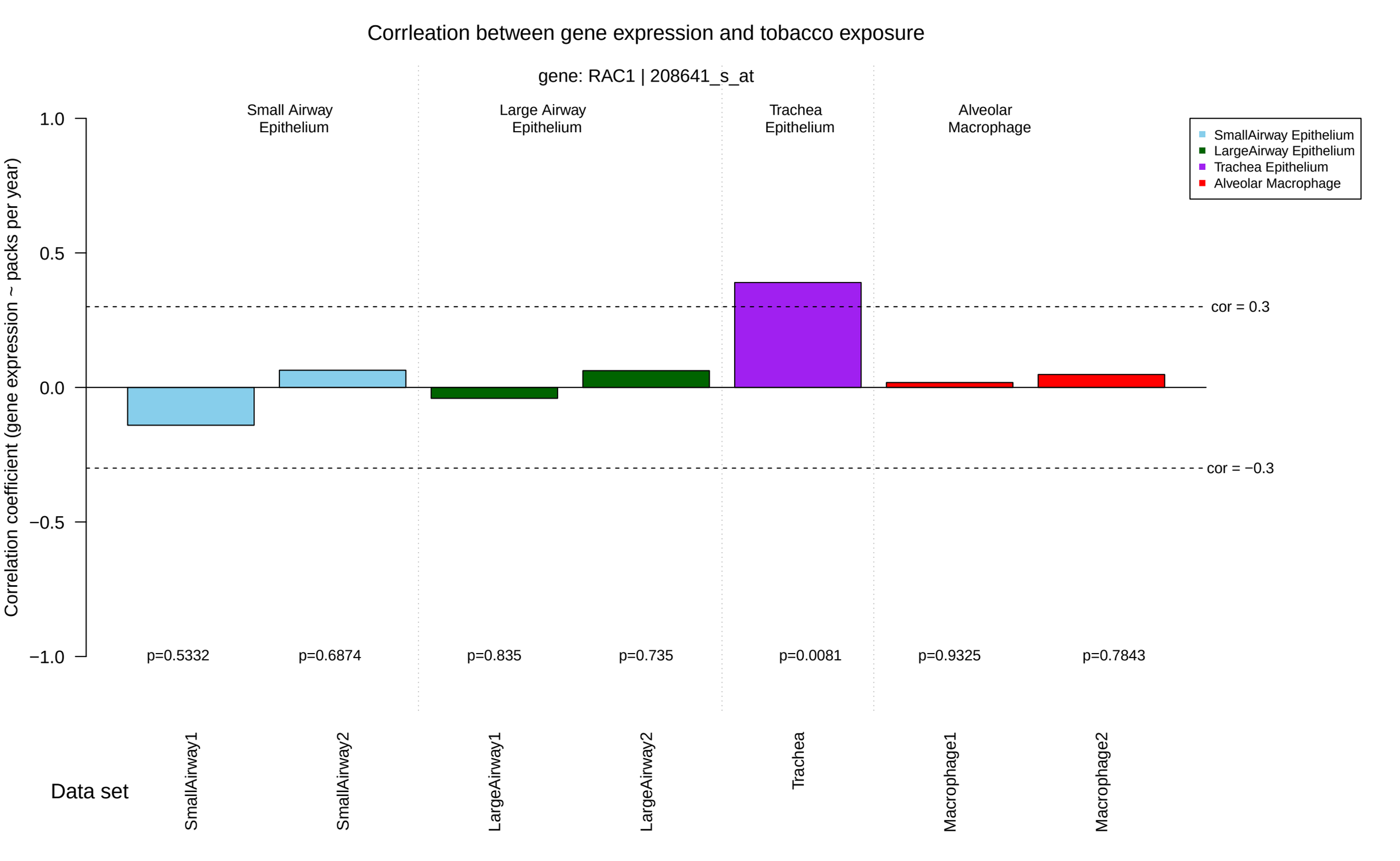

set: LargeAirway1 (p=0.3133)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5391)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5448)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9349)
RAC1 | 208641_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC1 | 208641_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5332
p=0.6874
p=0.835
p=0.735
p=0.0081
p=0.9325
p=0.7843
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2